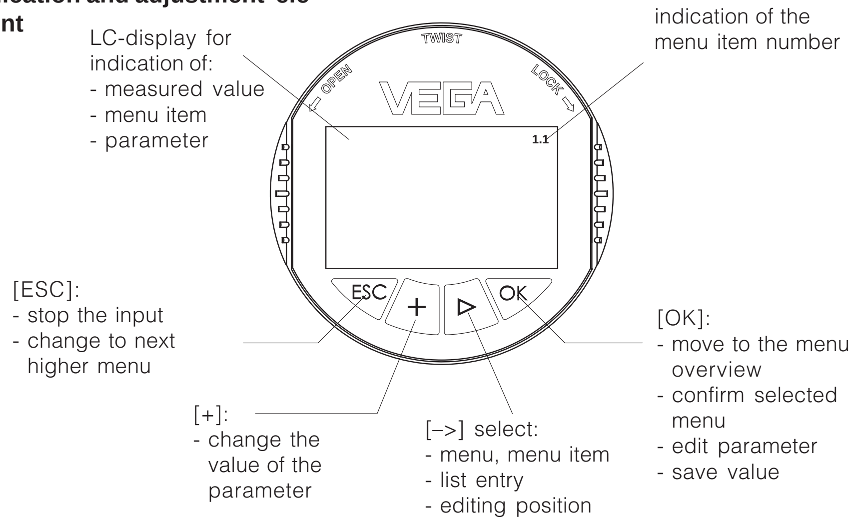
5 Set-up

5.1 Adjustment system

Indication and adjustment element

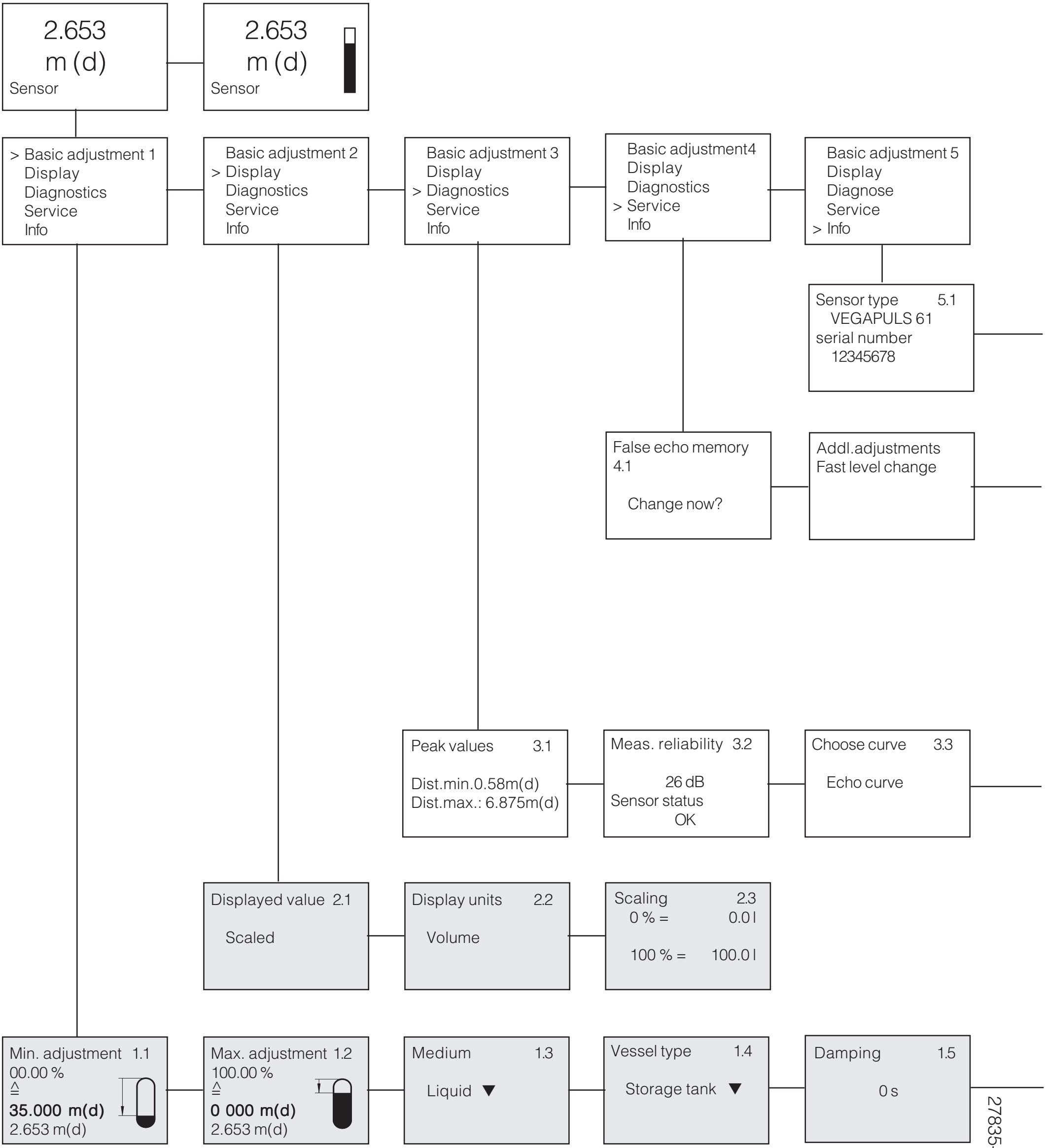


Adjustment system

The sensor is adjusted via the four keys of the indicating and adjustment module PLICSCOM. The LC display indicates the individual menu items. The functions are shown in the above illustration. Approx. 10 minutes after the last pressing of a key, an automatic reset to measured value display is triggered. Any values not confirmed with [OK] are not saved.

5.2 Menu schematic

Example of a 4 … 20 mA/HART® instrument



▼ Selection in menu item

[ESC]
[–>]
[OK]

With these keys you move in the menu field

Date of manufacture
5.2
26. September 2003
Software version
3.xx.xx

Date of last change
5.3
using PC
26. September 2003

Sensor details
Display now?

Current output 4.3
Output mode4-20mA
Failure mode<3.6mA
Min. current 3.8mA

Simulation 4.4
Start
simulation? ▼

Reset 4.5
Select
reset? ▼

Units of measure-
ment 4.6
m (d) ▼

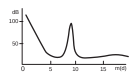
Language 4.7
Deutsch ▼

HART® operation
mode 4.8
Standard
Address 0

PIN 4.9
Enable?

Copy sensor data4.9
Copy sensor data?

Echo curve 3.4

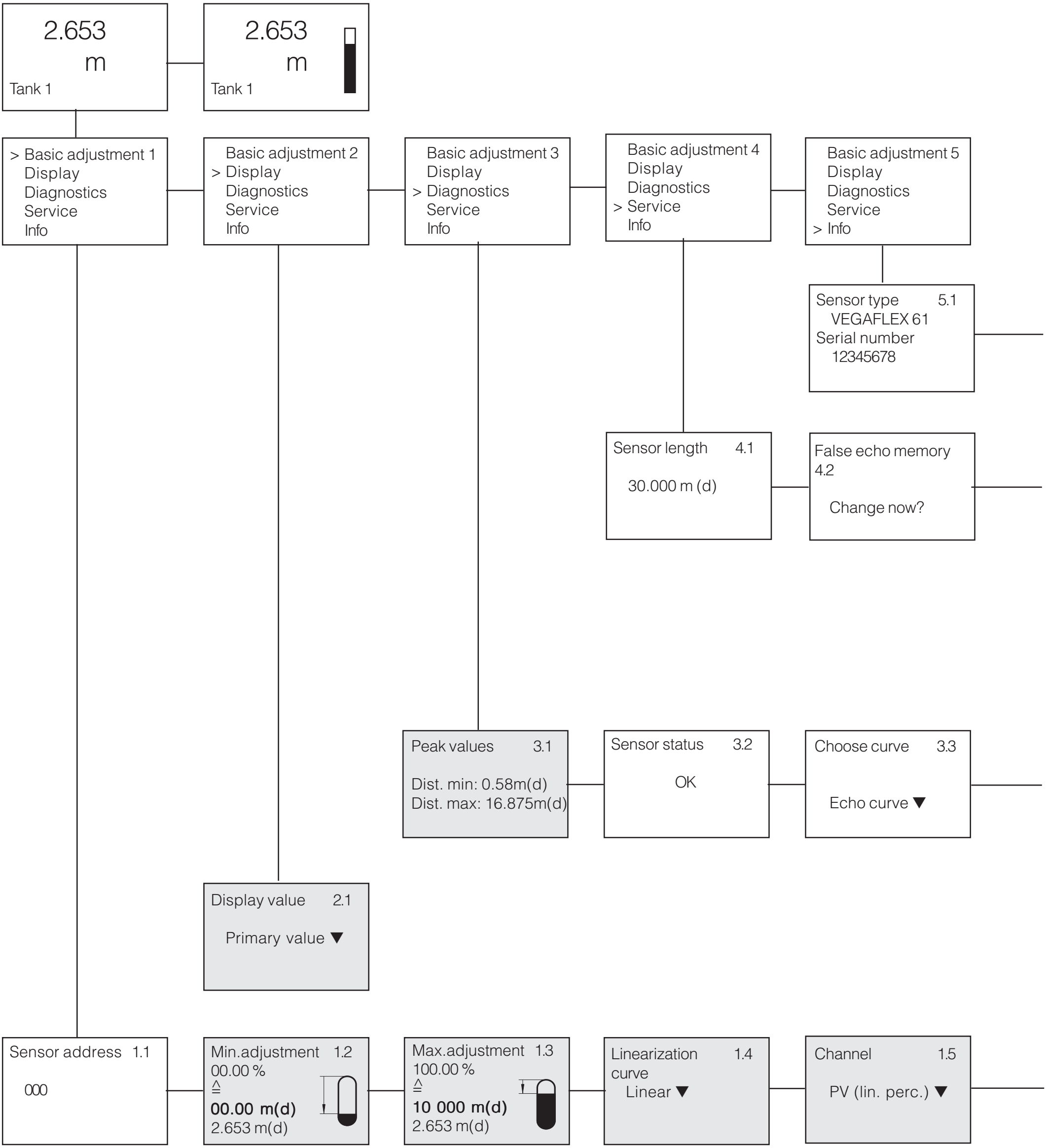


Linearization
curve1.6
Linear ▼

Sensor tag 1.7
Sensor

- This example refers to the VEGAPULS 61. You will find the instrument-specific menu schematic in the operating instructions manual of the respective instrument.
- The parameters in the menu items (highlighted in grey) are reset to default values with the reset function

Example of a Profibus PA instrument



27835-EN-031023

▼ Selection in menu item

[ESC]
[–>]
[OK]

With these keys you move in the menu field

Date of manufacture 5.2
26. September 2003
Software version
3.xx.xx

Date of last change 5.3
using PC
26. September 2003

Sensor details 5.4
Display now?

Additional PA 4.3
value
Secondary ▼
value 1

Out-Scale-Units 4.4
Volume ▼
l ▼

PV-/Out-Scale 4.5
100.00 lin %
≙ 0.0 l
0.00 lin %
≙ 100.0 l

Simulation 4.6
Start simulation ?
▼

Reset 4.7
Select reset?
▼

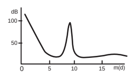
Units of measurement 4.8
m (d)

Language 4.9
Deutsch ▼

Copy sensor data 4.10
Copy sensor data ?

PIN 4.11
Disable ?

Echo curve 3.4



Damping 1.6
1 s

Sensor tag 1.7
Sensor

- This example refers to the VEGAFLEX 61. You will find the instrument-specific menu schematic in the operating instructions manual of the respective instrument.
- The parameters in the menu items (highlighted in grey) are reset to default values with the reset function

5.3 General functions

In the operating instructions manual of each instrument you will find the most important steps for quick set-up as well as a description of all sensor-specific functions.

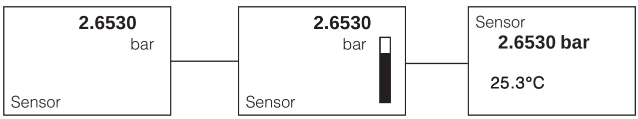
The general functions are described in this section. The scope of the functions of PLICSCOM is determined by the sensor and corresponds to the actual software version of the sensor.

The menu item number will differ depending on the instrument type and signal output.

Measured value display

In the measured value display you can select different presentations of the measured value with [–>]. From each of these presentations you reach the menu overview with [OK]. With [ESC] you move from the menu overview to the measured value presentation.

Presentation 1 | Presentation 2 | Presentation 3[1]



Indication of:

Presentation 1	Presentation 2	Presentation 3
– level/pressure as digital value – sensor tag	– level/pressure as digital value and analogue – sensor tag	– sensor tag – level/pressure as digital value – temperature value

Menu overview

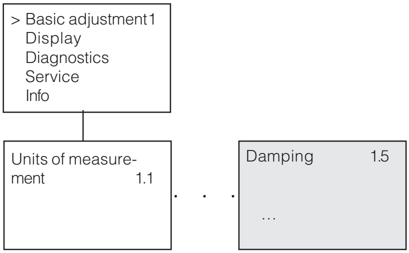
In the menu overview you select the requested menu with [–>] and open it with [OK]. The individual menu items are then at your disposal.

> Basic adjustment1
Display
Diagnostics
Service
Info

[1] Only with VEGABAR.

Damping

To damp pressure shocks and level fluctuations, you can set in this menu item an integration time of 0 … 999 s.



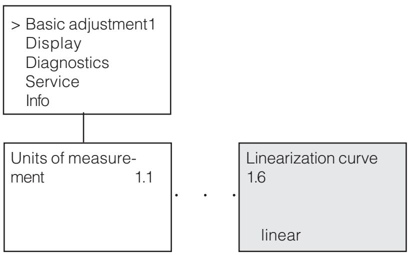
Linearisation curve

In this menu item you select the linearisation curve:

- linear
- cylindrical tank
- spherical tank
- user programmable.

User programmable means:
Switching on a linearisation curve created by the PC and PACT*ware*™.

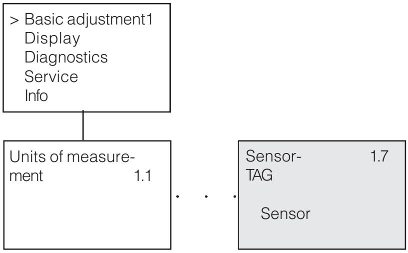
The linearisation curve establishes a correlation between height and volume. It takes the vessel geometry into account to provide correct measured value display and current output.



Edit sensor-TAG

In the menu item “Sensor-TAG” you can edit a twelve digit measurement loop name. The list of characters includes:

- letters A … Z
- numbers 0 … 9
- special characters _ ,+, -, /, –.



Peak values

Min. and max. measured values are stored in the sensor. The values are displayed in the menu item “Peak values“.

- Min. and max. distance in m (d): VEGAPULS, VEGASON, VEGAFLEX
- Min. and max. pressure: VEGABAR [1)]
- Min. and max. temperature: VEGABAR and VEGASON

Measuring accuracy

With VEGAPULS and VEGASON, the measurement can be affected by the prevailing process conditions. In this menu item, the measurement reliability of the level echo is displayed in dB. Measurement reliability is represented by signal intensity minus noise. The larger the value, the more reliable the measurement.

Instrument status

Display indicates „OK“ or flashing fault signal, e.g. „E013“. The error is also displayed in clear text in the measured value display.

Curve selection

With VEGAPULS, VEGASON and VEGAFLEX the following curves can be selected:

- echo curve
- false echo curve [2)]
- trend curve.

1) Pressure: -50 … +150 % of the nominal range; temperature -50 … +150°C.
2) Not with VEGAFLEX.

27835-EN-031023

The echo curve represents the echoes with signal strength in dB (VEGAPULS) resp. in V (VEGAFLEX) above the scaled distance. This enables a rough evaluation of the measurement.

The false echo curve represents the stored false echoes (see menu „Service“) of the empty vessel with signal strength in dB above the measuring range.

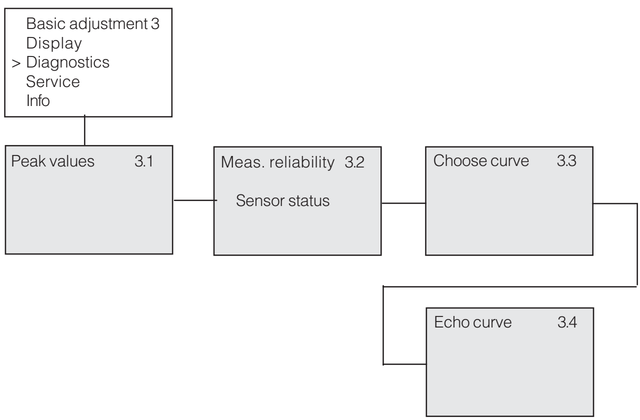
Curve presentation

A comparison of both curves gives more precise information about measurement reliability. The selected curve is continuously updated. By pushing the „OK“ key, a submenu with zoom function is opened. The echo and false echo curve enables:

- „X-Zoom“ zoom function for the distance
- „Y-Zoom“ 2, 5, and 10 fold increase of the signal in dB
- „Unzoom“ reset of the presentation to nominal range at normal magnification.

The trend echo curve enables:

- „X-Zoom“ resolution in minutes, hours or days
- „Stop/Start“ interruption of a running recording or beginning of a new recording
- „Unzoom“ reset of the resolution to minutes.



1) Not with VEGAFLEX.

Simulation of measured values

In this menu item you can simulate individual level or pressure values via the current output. By doing this, you can test, e.g. connected indicating instruments or the input card of the control system.

The following simulation units are available:

- percent
- current
- pressure (only with VEGABAR)
- distance (only with VEGAPULS and VEGAFLEX).

With Profibus PA sensors, the selection of the simulated value is carried out via the „channel" in the menu item „Basic adjustment".

This is how you start the simulation:

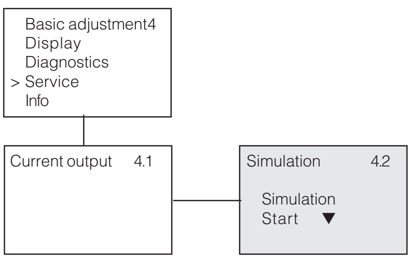
1 Push [OK]
2 Select the desired simulation unit with [–>] and confirm with [OK]
3 Set the desired number value with [+] and [–>]
4 Push [OK].

The simulation runs, i.e. the current 4 … 20 mA is thus outputted acc. to the simulation value.

To terminate the simulation:

–> Push [ESC]

10 min. after the last pressing of a key, the simulation is automatically terminated.



27835-EN-031023